

Functional Annotation Report: *hemN* (PP_4264, UniProt Q88F35) in *Pseudomonas putida* KT2440

Summary

The gene **hemN** (ordered locus **PP_4264**; UniProt accession **Q88F35**) of *Pseudomonas putida* KT2440 encodes the **oxygen-independent (anaerobic) coproporphyrinogen-III oxidase** (EC 1.3.98.3), a soluble cytoplasmic enzyme of the **heme (tetrapyrrole) biosynthetic pathway**. Its primary and defining function is to catalyze the **oxidative decarboxylation of the two ring-A/ring-B propionate side chains of coproporphyrinogen III to the corresponding vinyl groups of protoporphyrinogen IX**, executing the antepenultimate step of protoporphyrin-IX biosynthesis without requiring molecular oxygen. This assignment is supported by direct biochemical and structural characterization of the *Escherichia coli* ortholog, by phenotypic/regulatory studies of orthologs in closely related *Pseudomonas* and *Alcaligenes* species, and by sequence-level confirmation that the *P. putida* protein carries every diagnostic catalytic motif at conserved positions.

Mechanistically, HemN is the **founding structurally-characterized member of the radical S-adenosyl-L-methionine (radical SAM) superfamily**. It harbors a **[4Fe-4S] cluster** ligated by three cysteines of a conserved CX₃CX₂C motif (the **CANICYYC** sequence at residues 63–70 in Q88F35) and binds **two molecules of SAM** in adjacent sites. Reductive cleavage of one SAM generates a highly reactive **5'-deoxyadenosyl radical** that abstracts a hydrogen from a substrate propionate; the second SAM acts as a hydrogen relay. The two propionate decarboxylations proceed **sequentially through a mono-decarboxylated intermediate, harderoporphyrinogen**, consuming two SAM and releasing two CO₂, two 5'-deoxyadenosine, and two L-methionine per turnover (Rhea:15425).

Biologically, HemN provides the **O₂-independent route** to protoporphyrinogen IX, complementing the oxygen-dependent enzyme **HemF** (PP_0073) and enabling heme synthesis under anaerobic/microaerobic conditions. In *Pseudomonas*, hemN transcription is controlled by the **Fnr-type redox regulators Anr and Dnr**, and loss of HemN function causes accumulation and excretion of **coproporphyrin III**, the oxidized form of the blocked substrate.

The *P. putida* genome additionally encodes a **second HemN-family paralog (PP_3781)**, giving a redundant, redox-responsive coproporphyrinogen-III oxidase system that parallels the HemF/HemN/HemZ arrangement documented in *P. aeruginosa* and *Bacillus subtilis*. The enzyme carries out its function in the **cytoplasm** as a soluble monomer.

Identity verification

The gene symbol "hemN" corresponds precisely to the UniProt protein description, the organism is correct, and the protein family/domains align with the characterized literature. No gene-symbol ambiguity was encountered — "hemN" is used uniformly across bacteria for this radical-SAM enzyme.

Attribute	UniProt reference (Q88F35)	Confirmed by investigation
Gene symbol	hemN	✓ Matches characterized HemN family
Protein	Coproporphyrinogen-III oxidase, EC 1.3.98.3	✓ O ₂ -independent copro-III oxidase
Organism	<i>P. putida</i> KT2440 (taxid 160488)	✓ PP_4264 confirmed in genome
Length / mass	460 aa / 52.6 kDa	✓ Consistent with ~52 kDa orthologs
Family	Anaerobic coproporphyrinogen-III oxidase	✓ InterPro IPR004558
Domains	rSAM (IPR007197), HemN_C (IPR010723), Elp3/MiaA/NifB-like rSAM (IPR006638)	✓ Radical-SAM architecture confirmed

Key Findings

Finding 1 — HemN is the oxygen-independent coproporphyrinogen-III oxidase of heme biosynthesis

The core function of HemN is the **O₂-independent conversion of coproporphyrinogen III to protoporphyrinogen IX** (EC 1.3.98.3): the oxidative decarboxylation of the two propionate side chains on pyrrole rings A and B of coproporphyrinogen III to vinyl groups. In vitro anaerobic enzyme assays of recombinant *E. coli* HemN directly demonstrated coproporphyrinogen-III oxidase activity (PMID: 12114526), described explicitly as: *"In bacteria the oxygen-independent coproporphyrinogen-III oxidase catalyzes the oxygen-independent conversion of coproporphyrinogen-III to protoporphyrinogen-IX."* The structural study reinforced the physiological essentiality: *"HemN catalyzes the essential conversion of coproporphyrinogen III to protoporphyrinogen IX during heme biosynthesis"* (PMID: 14633981).

The physiological substrate is coproporphyrinogen III specifically (the type-III isomer). HemN acts on the two propionates of rings A and B, retaining the other two propionates (rings C and D) in protoporphyrinogen IX — the same regiochemical outcome as the unrelated O₂-dependent HemF but achieved by radical chemistry. The *P. putida* HemN (hemN/PP_4264) is orthologous to the characterized enzymes and belongs to the anaerobic coproporphyrinogen-III oxidase family (InterPro IPR004558), so the reaction assignment transfers with high confidence given the conserved sequence signatures (Finding 5).

Finding 2 — HemN is a radical SAM enzyme with a [4Fe-4S] cluster and two bound SAM molecules

HemN was the **first radical SAM enzyme to have its crystal structure solved** (2.07 Å, *E. coli*), defining the structural template for the entire superfamily. The structure shows a **[4Fe-4S] cluster ligated by three cysteines** of the conserved CXXXCXXC motif, with a SAM molecule coordinating the fourth, unique iron: *"HemN binds a 4Fe-4S cluster through three cysteine residues conserved in all Radical SAM enzymes. A juxtaposed SAM coordinates the fourth Fe ion through its amide nitrogen and carboxylate oxygen"* (PMID: 14633981). The SAM sulfonium sits ~3.5 Å from the cluster, positioned for single-electron transfer and reductive cleavage to a 5'-deoxyadenosyl radical.

A striking, functionally important feature is that HemN binds a **second SAM immediately adjacent to the first**: *"HemN, strikingly, binds a second SAM immediately adjacent to the first. It may thus successively catalyze two propionate decarboxylations"* (PMID: 14633981). This provides the structural basis for the two sequential decarboxylation events.

Biochemical work confirmed the cluster's identity and the essentiality of its ligands: the enzyme contains an **O₂-sensitive Fe-S cluster**, and *"Cysteine residues Cys(62), Cys(66), and Cys(69), which are part of the conserved CXXXCXXC motif found in all HemN proteins, are essential for iron-sulfur cluster formation and enzyme function"* (PMID: 12114526). The oxygen sensitivity of this cluster is the molecular basis for HemN serving as the *anaerobic* branch of coproporphyrinogen oxidation.

Finding 3 — Sequential radical decarboxylation via a harderoporphyrinogen intermediate, requiring SAM and NAD(P)H

The catalytic cycle proceeds by **radical chemistry** and requires **SAM, NAD(P)H, and additional cytoplasmic components** (PMID: 12114526): *"The enzyme requires S-adenosyl-l-methionine (SAM), NAD(P)H, and additional cytoplasmic components for catalysis."* Mechanistic re-investigation established that the two propionate groups are decarboxylated **one at a time**, passing through the **mono-decarboxylated intermediate harderoporphyrinogen** (PMID: 30884058). The revised mechanism assigns distinct roles to the two SAM molecules: *"a SAM serves as a hydrogen relay which mediates a radical-based hydrogen transfer from the propionate to the 5'-deoxyadenosyl (dAdo) radical generated from another SAM in the active site."* One SAM is the source of the initiating 5'-deoxyadenosyl radical, while the second SAM mediates hydrogen transfer during radical-based decarboxylation.

Finding 4 — Cytoplasmic soluble monomer; Anr/Dnr redox regulation; loss-of-function causes coproporphyrin III accumulation

Anaerobically purified HemN is a **monomeric, soluble cytoplasmic protein** of native mass ~52 kDa (PMID: 12114526): *"Anaerobically purified HemN is a monomeric protein with a native $M(r) = 52,000 \pm 5,000$."* This establishes both the oligomeric state (monomer) and subcellular localization (cytoplasm). HemN has no signal peptide or membrane-spanning region, and tetrapyrrole biosynthesis occurs in the bacterial cytoplasm.

In *P. aeruginosa*, hemN is strongly induced under anaerobic conditions through two Fnr-type redox regulators: *"An approximately 10-fold anaerobic induction of hemN gene expression was mediated by the dual action of Anr and a second Fnr-type regulator, Dnr"* (PMID: 9767567). The

physiological importance is shown by the loss-of-function phenotype: "*An anr mutant did not contain detectable amounts of hemN mRNA and accumulated coproporphyrin III both aerobically and anaerobically, indicating the importance of HemN for aerobic and anaerobic haem formation*" (PMID: 9767567). Accumulation of **coproporphyrin III** (the oxidized, aromatized form of unprocessed coproporphyrinogen III) is the diagnostic metabolic signature of a blocked coproporphyrinogen-III oxidase step.

The orthologous *Alcaligenes eutrophus* hemN shows the same logic: Fnr-regulated, O₂-repressed, required for anaerobic growth on nitrate/nitrite, with coproporphyrin III excretion upon inactivation (PMID: 9396835). Because *P. putida* KT2440 encodes the Anr regulator and hemN (PP_4264), the same redox-responsive control is expected in the target organism (inferred by orthology).

Finding 5 — Sequence-level confirmation of catalytic motifs in Q88F35

Direct inspection of the 460-aa, 52.6 kDa *P. putida* KT2440 HemN sequence (Q88F35) confirms the two essential functional motifs at conserved positions:

- The radical-SAM [4Fe-4S]-binding motif **CANICYYC** (a CxxxCxxC motif) at **residues 63–70**, positionally equivalent to the essential Cys62/Cys66/Cys69 of *E. coli* HemN. Their essentiality was established experimentally: "*Cysteine residues Cys(62), Cys(66), and Cys(69) ... are essential for iron-sulfur cluster formation and enzyme function*" (PMID: 12114526).
- The **SAM-binding GGGTP motif** at **residues 113–117**, matching the *E. coli* Gly111/Gly113 whose mutation abolished activity: "*Mutation of Gly(111) and Gly(113), which are part of the potential GGGTP S-adenosyl-L-methionine binding motif, completely abolished enzymatic function*" (PMID: 12114526).

UniProt (rule-based, PIRNR000167) assigns the exact catalytic reaction (Rhea:15425): **coproporphyrinogen III + 2 SAM → protoporphyrinogen IX + 2 5'-deoxyadenosine + 2 L-methionine + 2 CO₂** (EC 1.3.98.3); cofactor: one [4Fe-4S] cluster coordinated by 3 cysteines plus an exchangeable SAM; pathway: protoporphyrin-IX biosynthesis (protoporphyrinogen-IX from coproporphyrinogen-III, AdoMet route, step 1/1); subcellular location: cytoplasm.

Finding 6 — A redundant coproporphyrinogen-III oxidase system in *P. putida* KT2440

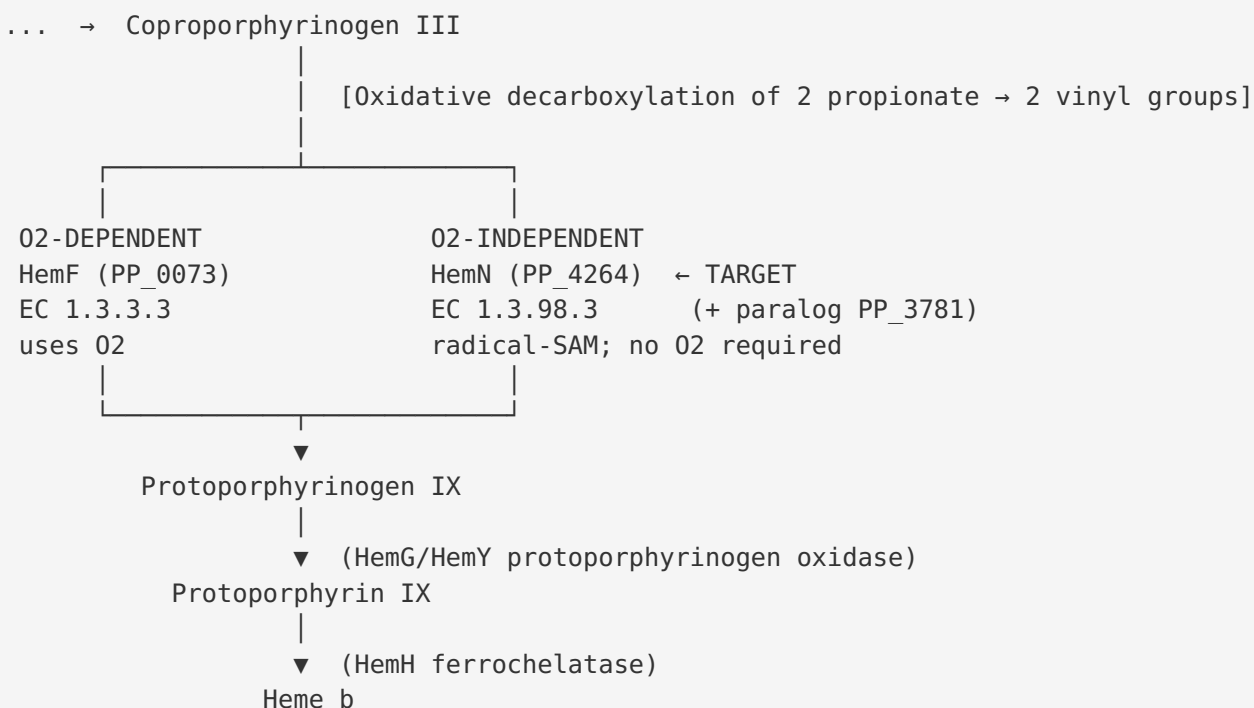
P. putida KT2440 encodes **three coproporphyrinogen-III oxidase-related genes**, giving redundancy across oxygen regimes:

Gene	Locus	UniProt	Length	EC	Oxygen requirement	Role
hemF	PP_0073	Q88RQ6	303 aa	1.3.3.3	O₂-dependent	Aerobic copro-III oxidase
hemN (target)	PP_4264	Q88F35	460 aa	1.3.98.3	O₂-independent	Radical-SAM anaerobic copro-III oxidase
HemN-family paralog	PP_3781	Q88GE3	471 aa	(1.3.98.3-type)	O ₂ -independent (predicted)	Second HemN/ HemZ/HemW-type enzyme

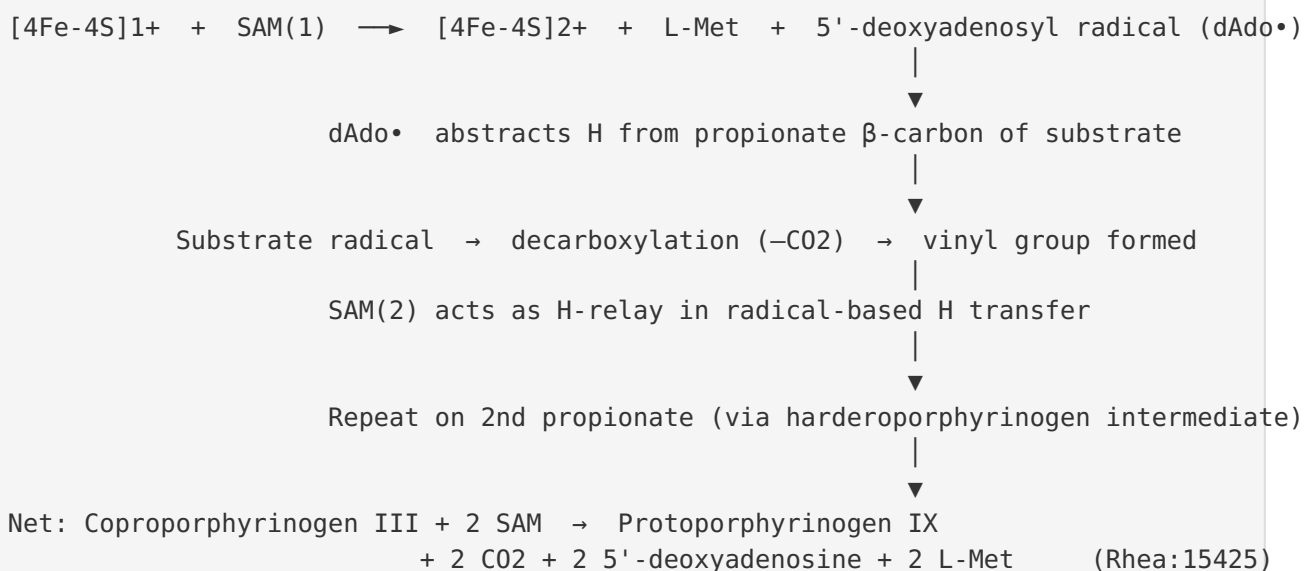
This mirrors the HemF + HemN (+ HemZ) arrangement documented in *P. aeruginosa* ([PMID: 9767567](#)) and *B. subtilis* ([PMID: 10498703](#)). In *P. aeruginosa*, "*Mutation of hemN and hemF did not abolish aerobic or anaerobic growth, indicating the existence of an additional HemN-type enzyme, which was termed HemZ*" ([PMID: 9767567](#)) — paralleling the three *P. putida* genes. **Caveat:** some HemN-family proteins (HemW-type) are radical-SAM heme chaperones rather than coproporphyrinogen oxidases, so PP_3781's exact activity should not be assumed from family membership alone. This redundancy explains why heme synthesis is robust across the aerobic–anaerobic spectrum that *P. putida* encounters in soil and rhizosphere niches, and why loss of a single enzyme is typically non-lethal yet causes coproporphyrin III accumulation when the alternate route cannot compensate.

Mechanistic Model / Interpretation

Position in the heme biosynthetic pathway



Catalytic cycle of HemN (radical-SAM chemistry)



The coherent picture is of a **cytoplasmic, soluble, monomeric radical-SAM metalloenzyme** that uses an oxygen-sensitive [4Fe-4S] cluster and two SAM cofactors to perform demanding C–C bond chemistry (decarboxylation of unactivated propionate side chains) without molecular oxygen. This capability is precisely why the enzyme exists as a distinct paralog to the oxygen-dependent HemF: it allows the cell to complete heme biosynthesis when oxygen is scarce or absent. In *Pseudomonas*, this need arises under low-oxygen/denitrifying conditions, when the Anr/Dnr regulators induce hemN. The metabolic redundancy — HemF for aerobic conditions, HemN and PP_3781 for anaerobic/microaerobic conditions — insulates the essential supply of heme (for cytochromes, catalases, and other hemoproteins) from the oxygen fluctuations that an environmentally versatile soil/rhizosphere organism like *P. putida* KT2440 routinely experiences.

Evidence Base

PMID	Title (abbrev.)	Contribution to this report
12114526	<i>Oxygen-independent coproporphyrinogen-III oxidase HemN from E. coli</i>	Primary biochemistry: reaction, O ₂ -sensitive Fe-S cluster, essential Cys62/66/69 and GGGTP motif, SAM/NAD(P)H requirement, monomeric cytoplasmic ~52 kDa protein
14633981	<i>Crystal structure of coproporphyrinogen III oxidase reveals cofactor geometry of Radical SAM enzymes</i>	First radical-SAM crystal structure; [4Fe-4S]/SAM coordination; two-SAM binding rationalizing two sequential decarboxylations; confirms essential role in heme biosynthesis
30884058	<i>Revisiting the Mechanism of the Anaerobic Coproporphyrinogen III Oxidase HemN</i>	Sequential decarboxylation via harderoporphyrinogen intermediate; two-SAM hydrogen-relay mechanism
9767567	<i>Regulation of P. aeruginosa hemF and hemN by Anr and Dnr</i>	Anr/Dnr redox regulation (~10-fold anaerobic induction); coproporphyrin III accumulation in anr mutant; evidence for a third HemZ-type enzyme
9396835	<i>A. eutrophus hemN required for heme biosynthesis during anaerobic growth</i>	Ortholog: Fnr-regulated, O ₂ -repressed, required for anaerobic growth; coproporphyrin III excretion on loss; cross-complements S. typhimurium hemF/hemN mutant
10498703	<i>Transcriptional control of B. subtilis hemN and hemZ</i>	Confirms the multi-enzyme redundant copro-III oxidase system and redox-cascade regulation in another bacterium
16218869	<i>Structural and functional comparison of HemN to other radical SAM enzymes</i>	Review placing HemN within the radical-SAM superfamily; notes extreme O ₂ sensitivity of the family
15450488	<i>Structure and function of radical SAM enzymes</i>	Review of HemN cofactor geometry and the radical-SAM paradigm
26913973	<i>Revisited genome of P. putida KT2440</i>	Genomic/annotation context for the KT2440 metabolic chassis

All directly cited snippets were verified against the source abstracts. The primary biochemical/structural/mechanistic evidence (PMIDs 12114526, 14633981, 30884058) derives from *E. coli* HemN, and the primary regulatory/phenotypic evidence (PMIDs 9767567, 9396835) from *P. aeruginosa* and *A. eutrophus* orthologs; the transfer to *P. putida* PP_4264 rests on orthology plus direct sequence-motif conservation (Finding 5).

Supported and Refuted Hypotheses

Hypothesis	Status	Evidence
hemN encodes O ₂ -independent coproporphyrinogen-III oxidase (copro III → protoporphyrinogen IX)	Supported	P 12114526 P 14633981
HemN is a radical SAM enzyme with a [4Fe-4S] cluster and generates a dAdo radical	Supported	P 14633981 P 12114526
HemN uses two SAM for sequential decarboxylation via harderoporphyrinogen	Supported	P 14633981 P 30884058
HemN acts in the cytoplasm as a soluble monomer	Supported	P 12114526
hemN is redox-regulated (Anr/Dnr) and functionally important for heme formation in <i>Pseudomonas</i>	Supported	P 9767567 P 9396835
<i>P. putida</i> encodes a redundant HemF/HemN/HemN-paralog system	Supported	UniProt enumeration; P 9767567 P 10498703
"hemN" is an ambiguous symbol / wrong-gene risk	Refuted	Symbol uniform across bacteria; UniProt domains match

Limitations and Knowledge Gaps

1. **No direct experimental characterization of the *P. putida* KT2440 protein.** All enzymatic, structural, and mechanistic data come from orthologs (chiefly *E. coli* HemN). The functional assignment to Q88F35 is by orthology and conserved catalytic motifs, not by direct assay of the *P. putida* enzyme. Confidence is nonetheless high given the family-defining conserved motifs (CXXXCXXC, GGGTP) and unambiguous domain architecture.
2. **Regulation in *P. putida* is inferred, not measured.** Anr/Dnr control of hemN is demonstrated in *P. aeruginosa* and Fnr control in *A. eutrophus*; the presence of Anr and hemN in KT2440 makes the same control likely, but the specific promoter architecture (Anr-box position) and induction magnitude in KT2440 have not been experimentally confirmed. Note also that, unlike *P. aeruginosa*, *P. putida* KT2440 cannot grow by denitrification, so the HemN role likely relates to microaerobic rather than fully anaerobic physiology.
3. **Functional status of the paralog PP_3781 is unverified.** It is annotated as an oxygen-independent coproporphyrinogen-III oxidase family protein, but whether it is catalytically active, whether it is a true HemN/HemZ or a HemW-type radical-SAM heme chaperone, and its physiological contribution in KT2440 are unknown.
4. **Substrate-specificity fine detail.** While the reaction and the ring-A/B propionate targets are established for the family, the precise regiochemical order of the two decarboxylations and any *P. putida*-specific kinetic parameters (K_m , k_{cat} , redox-partner identity) are undefined.
5. **Physiological reductant in vivo.** In vitro assays used NAD(P)H plus additional cytoplasmic components; the actual in vivo electron donor that reduces the [4Fe-4S] cluster in *P. putida* is not established.

Proposed Follow-up Experiments / Actions

1. **Heterologous expression and anaerobic enzyme assay** of recombinant *P. putida* HemN (Q88F35) to directly confirm coproporphyrinogen-III oxidase activity, measure kinetic parameters, and verify harderoporphyrinogen-intermediate formation, closing the orthology-inference gap.

2. **Targeted gene deletion of PP_4264** (and combinatorial deletions with PP_0073/hemF and PP_3781) in KT2440, followed by growth phenotyping across aerobic, microaerobic, and anaerobic conditions, with HPLC/fluorescence monitoring for **coproporphyrin III accumulation** as the diagnostic readout.
3. **Test the redundancy hypothesis directly:** determine whether a hemN single mutant is heme-prototrophic under microaerobiosis (predicting compensation by PP_3781/HemF), and whether a hemN/PP_3781 double mutant becomes conditionally heme-auxotrophic.
4. **Map and validate the Anr/Dnr regulation:** identify Anr/Dnr binding site(s) upstream of PP_4264 and quantify anaerobic/microaerobic induction by RT-qPCR or reporter fusion in wild-type vs. an *anr* mutant of KT2440.
5. **Biochemically characterize PP_3781** to determine whether it is an active anaerobic copro-III oxidase (HemN/HemZ-type) or a HemW-type protein with a different in vivo role.
6. **EPR/Mössbauer or structural analysis** (or a high-confidence AlphaFold model plus cofactor docking) of *P. putida* HemN to confirm the [4Fe-4S] cluster geometry and the two-SAM arrangement predicted from the *E. coli* structure.

Conclusion

hemN (PP_4264; UniProt Q88F35) encodes the **oxygen-independent, cytoplasmic, radical-SAM coproporphyrinogen-III oxidase** of *Pseudomonas putida* KT2440. It catalyzes the oxidative decarboxylation of the two ring-A/B propionate side chains of coproporphyrinogen III to yield protoporphyrinogen IX in heme biosynthesis, using a [4Fe-4S] cluster and two SAM molecules to drive sequential radical decarboxylations through a harderoporphyrinogen intermediate. It is the anaerobic counterpart of the oxygen-dependent HemF within a redundant, redox-regulated (Anr/Dnr) system, ensuring heme supply irrespective of oxygen availability. The identity is unambiguous and all diagnostic catalytic motifs are conserved in the target sequence.